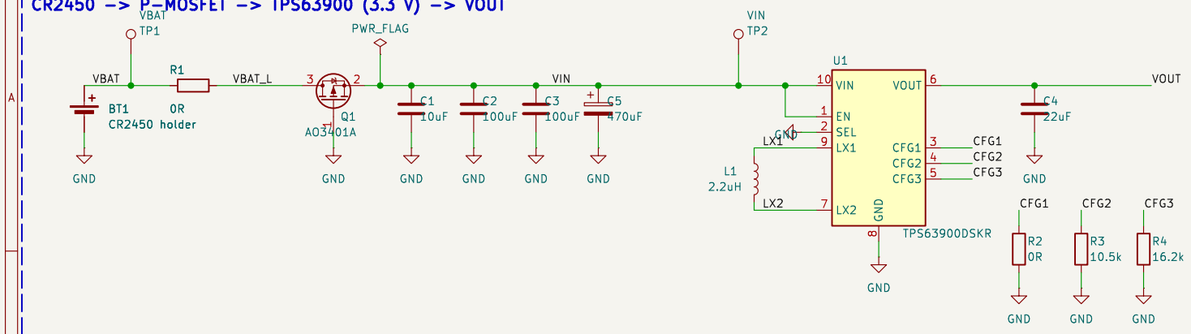
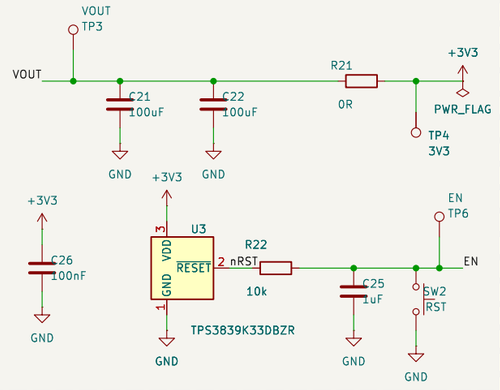
CR2450 -> P-MOSFET -> TPS63900 (3.3 V) -> VOUT

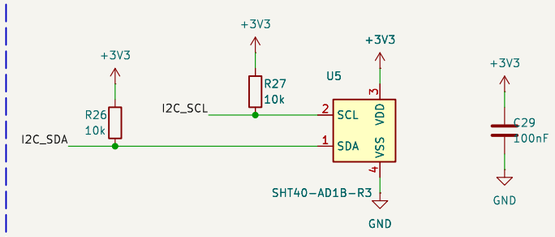


SEL = L: VO(1) = 3.3 V (RCFG3 16.2k, table 7-3). Input current limit 100 mA from RCFG1 0R / RCFG2 10.5k (table 7-2: 13.3k = 50 mA option). TI SLVSET3D, +-1 % resistors. C5: 470 uF electrolytic buffer.

3V3 RAIL, SUPERVISOR, RESET

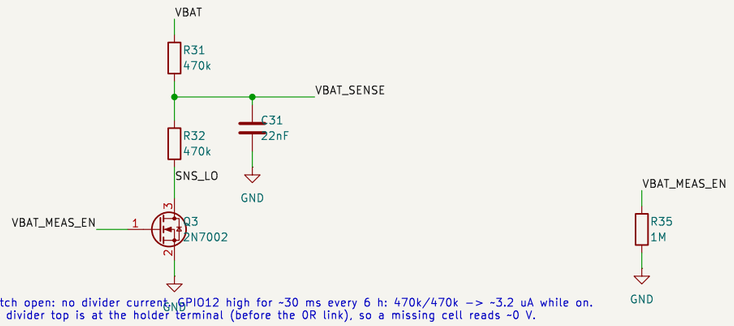


SHT40 (on a slotted tongue, away from converters and module)



No copper under the sensor except its pads; do not solder the die pad (SHT4x datasheet).

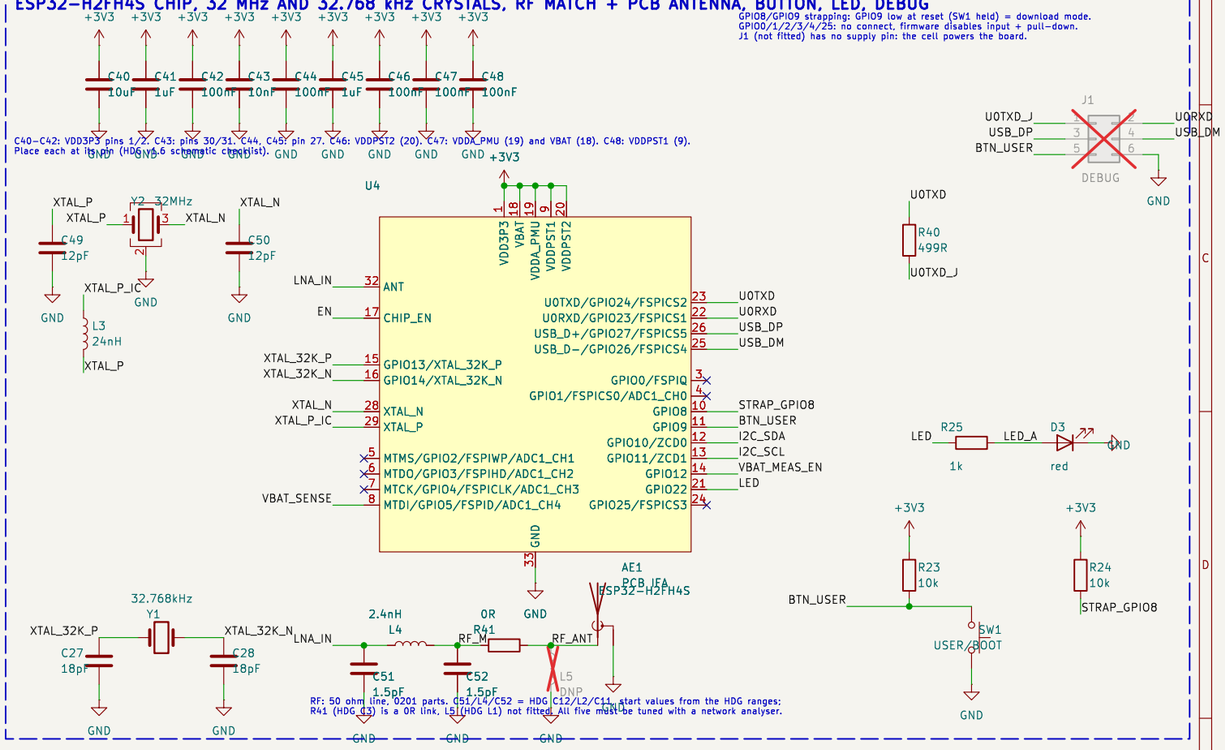
CELL VOLTAGE SENSE (switched 470k/470k divider, ADC1_CH4 on GPIO5)



Switch open: no divider current, GPIO12 high for ~30 ms every 6 h: 470k/470k -> ~3.2 uA while on. The divider top is at the holder terminal (before the 0R link), so a missing cell reads ~0 V.

ESP32-H2FH4S CHIP, 32 MHz AND 32.768 kHz CRYSTALS, RF MATCH + PCB ANTENNA, BUTTON, LED, DEBUG

GPIO8/GPIO9 strapping: GPIO9 low at reset (SW1 held) = download mode.
GPIO0/1/2/3/4/25: no connect, firmware disables input + pull-down.
J1 (not fitted) has no supply pin: the cell powers the board.



TEST PADS, MECHANICAL, NOTES



CR2450 ONLY — NOT RECHARGEABLE. No other supply can reach the rail: the cell is the only energy source.
Power-path decision and its limits: docs/decisions.md, docs/power_budget.md (ESTIMATE until measured).
Current measurement: remove R1 (cell) or R21 (load) and insert the ammeter.
U3 TPS3839K33: 2.93 V typ (2.857-2.974 V), push-pull -RESET, 200 ms delay; R22 keeps it from fighting SW2 / TP6 (EN).

Generated by hardware/scripts/gen_schematic.py from design.py — edit there
ERC/DRC reports: hardware/chip/reports
Power path and battery life figures are ESTIMATES (docs/power_budget.md)
ESP32-H2FH4S chip + PCB antenna, Matter over Thread, SHT40, 1 x CR2450, TPS63900 3.3 V

TH-SENSOR-H2C

Sheet: /
File: th_sensor_h2c.kicad_sch

Title: TH-SENSOR-H2C thermo-hygrometer (bare ESP32-H2 chip)

Size: A3	Date: 2026-11-06	**Rev: A**
KiCad E.D.A. 9.0.9		Id: 1/1